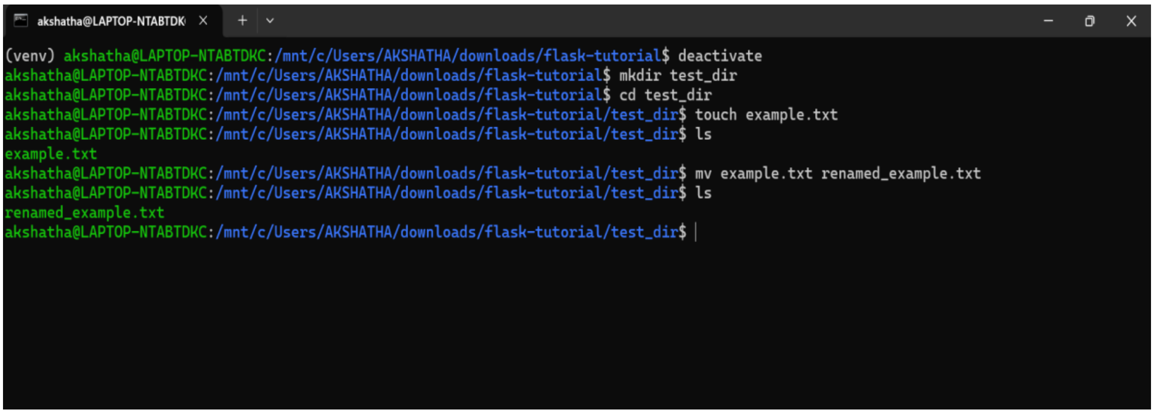
1. Creating and Renaming Files/Directories



```
(venv) akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ deactivate
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ mkdir test_dir
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ cd test_dir
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ touch example.txt
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ ls
example.txt
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ mv example.txt renamed_example.txt
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ ls
renamed_example.txt
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ |
```

mkdir test_dir creates a new directory named test_dir

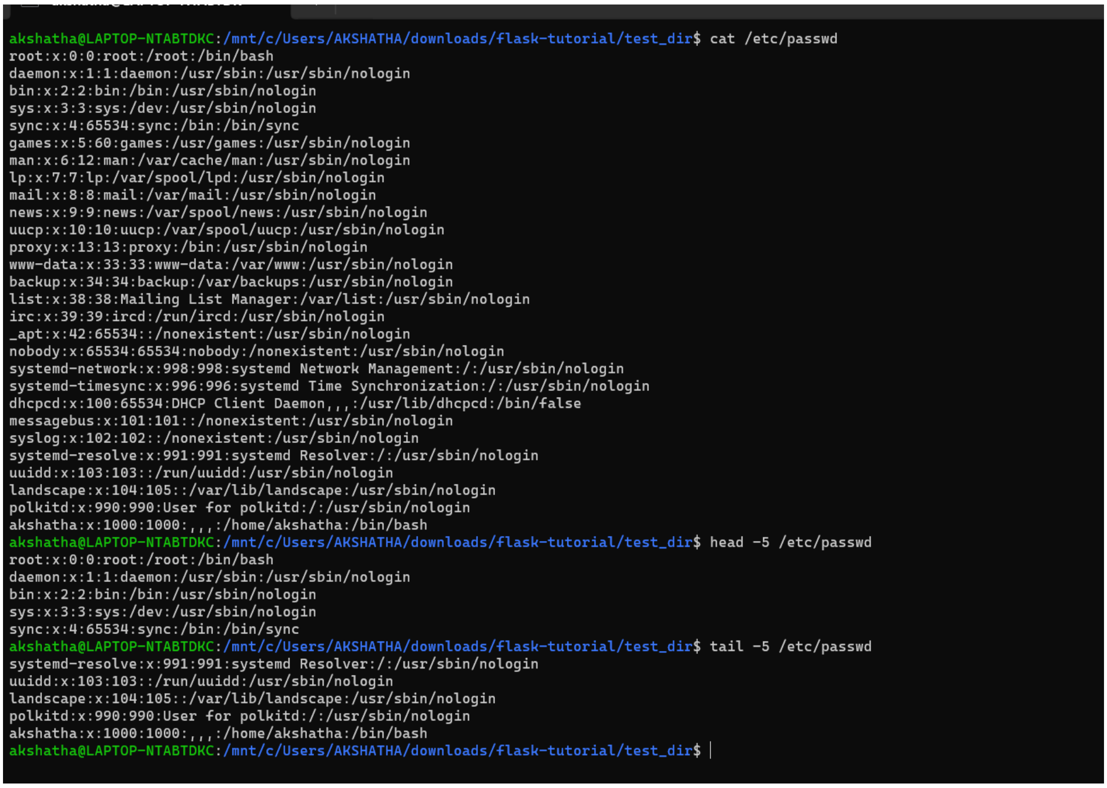
cd test_dir changes the current directory to test_dir

touch example.txt creates an empty file named example.txt

mv example.txt renamed_example.txt renames file from example.txt to renamed_example.txt

ls lists the files in the directory to verify that the file has been renamed successfully

2. Viewing File Contents



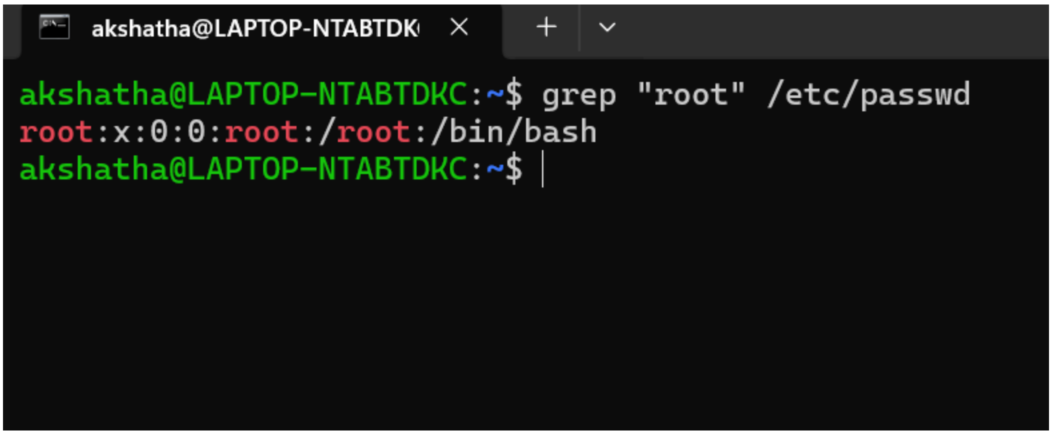
```
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
akshatha:x:1000:1000:,,,:/home/akshatha:/bin/bash
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ head -5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ tail -5 /etc/passwd
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
akshatha:x:1000:1000:,,,:/home/akshatha:/bin/bash
akshatha@LAPTOP-NTABTDKC:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ |
```

cat /etc/passwd displays the complete contents of the /etc/passwd file

head -n 5 /etc/passwd displays the first 5 lines of the file

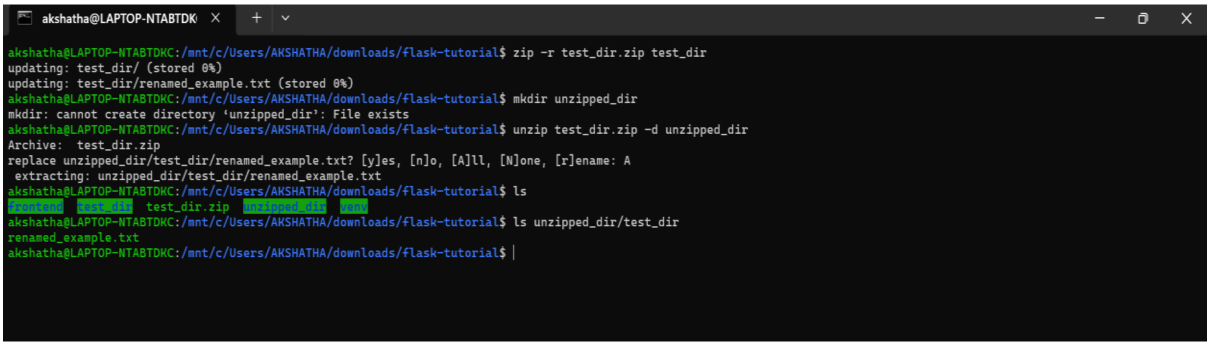
tail -n 5 /etc/passwd displays the last 5 lines of the file

3.Searching for Patterns



grep “root” /etc/passwd searches the /etc/passwd file for all lines containing the word “root”

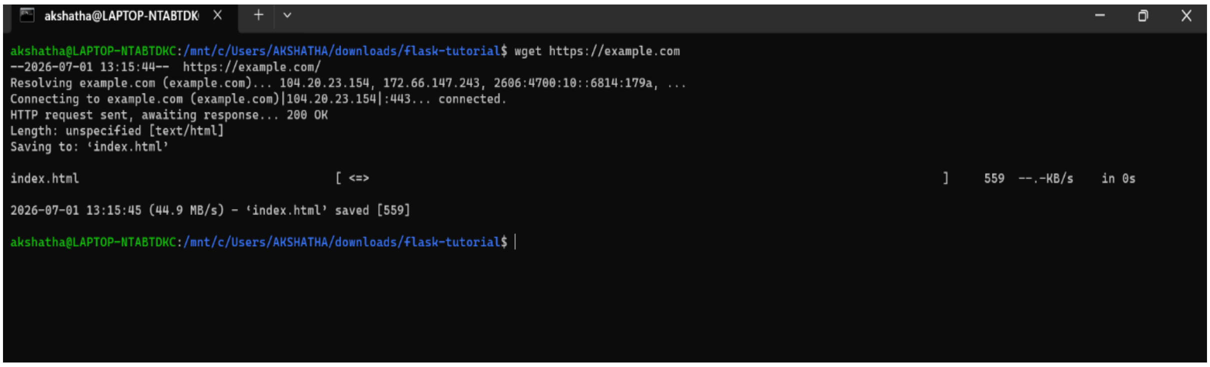
4. Zipping and Unzipping



zip -r test_dir.zip test_dir compresses the test_dir directory and all its contents into a ZIP file named test_dir.zip The -r option means recursive, so it includes all files and subdirectories

unzip test_dir.zip -d unzipped_dir extracts the contents of the ZIP file into a new directory named unzipped_dir

5. Downloading Files



wget is a command-line utility used to download files from the internet

In this task, wget https://example.com downloads the homepage of the website example.com and saves it as index.html in the current directory

6. Changing Permissions

touch secure.txt creates an empty file named secure.txt.

chmod 444 secure.txt changes the file permissions to read-only for the owner, group, and others.

ls -l secure.txt displays the file permissions to verify that the changes were applied successfully.

7. Working with Environment Variables

export MY_VAR="Hello, Linux!" creates an environment variable named MY_VAR and assigns it the value "Hello, Linux!".

echo $MY_VAR displays the value of the environment variable.